Name _____ Period _____ Date _____

DNA Structure & Replication

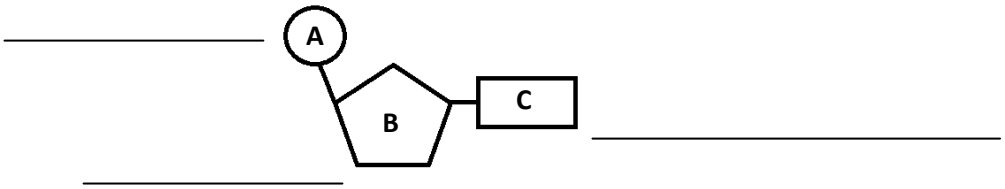
Directions: Answer each question, complete each statement, or label each diagram as indicated. Use your textbook, notes, or other appropriate resources. If using your text, refer to pages _____.

1. What is the full name of **DNA**? _____

2. Segments of DNA which code for specific proteins are called _____.

3. DNA may coil and condense into visible structures called _____.

4. DNA is a polymer of **nucleotides**. Label the parts of the nucleotide shown below.

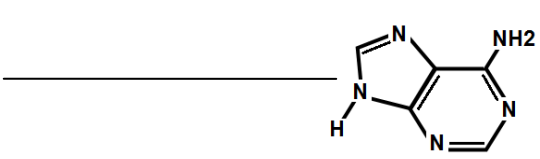


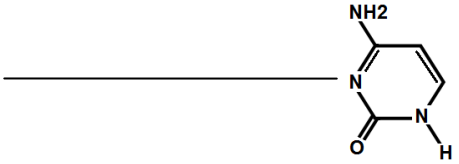
5. Four **nitrogenous bases** are found in **DNA**.

   a) Which two nitrogenous bases are **purines**? _____ _____

   b) Which two nitrogenous bases are **pyrimidines**? _____ _____

6. Classify the nitrogenous bases below as either a **purine** or a **pyrimidine**.





7. Nitrogenous bases form pairs in specific combinations. Identify these pairs in the spaces below.

   a) _____ always pairs with _____

   b) _____ always pairs with _____

8. What chemical **bond** holds the nitrogenous base pairs together? _____

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Name _____ Period _____ Date _____

DNA Structure & Replication, *Continued*

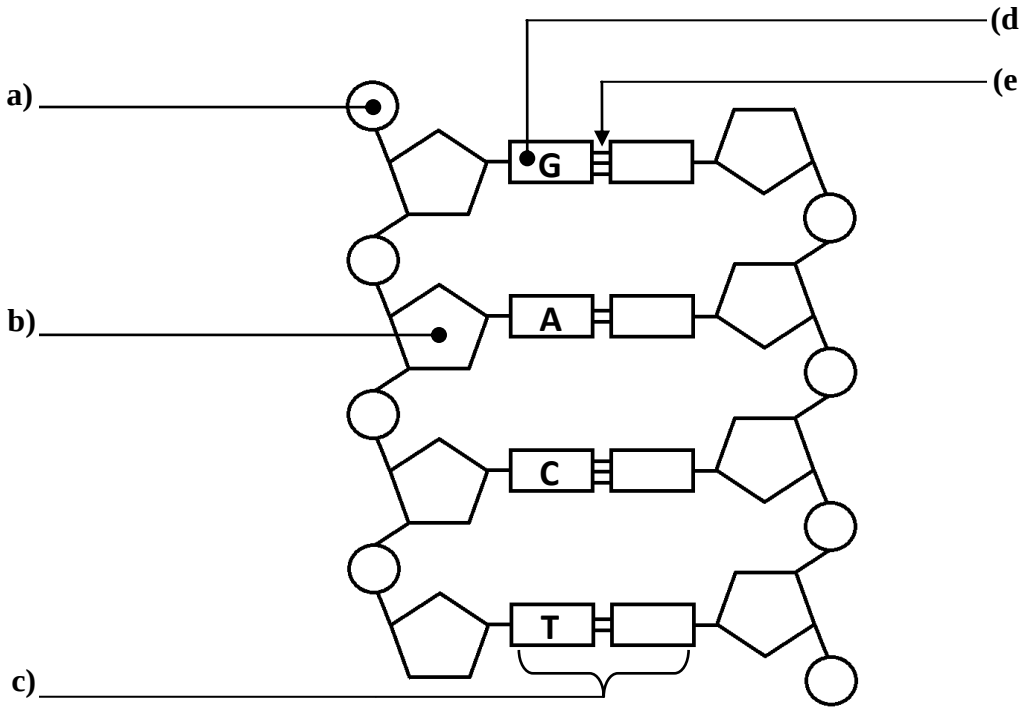
9. James Watson and Francis Crick, with the help of Rosalind Franklin, determined that DNA is in the shape of a _____ _____.

10. The shape of DNA can be compared to a twisted ladder.

    a) The sides of the ladder would be the _____ and _____.

    b) The rungs or steps of the ladder would be the _____.

11. Label the DNA diagram shown below using the follow terms: **base pair, deoxyribose**, **hydrogen bond**, **nitrogenous base**, and **phosphate group**. Then, fill in the missing nitrogenous bases.



12. DNA undergoes **replication** before a cell divides by mitosis or meiosis.

    a) What enzyme **"**unzips" the DNA molecule? _____

    b) What enzyme builds the new DNA strands using the original strands as templates?

    _____

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Name _____ Period _____ Date _____

DNA Structure & Replication, *Continued*

13. Number the following steps of **DNA replication** in the correct order (1, 2, 3, …)

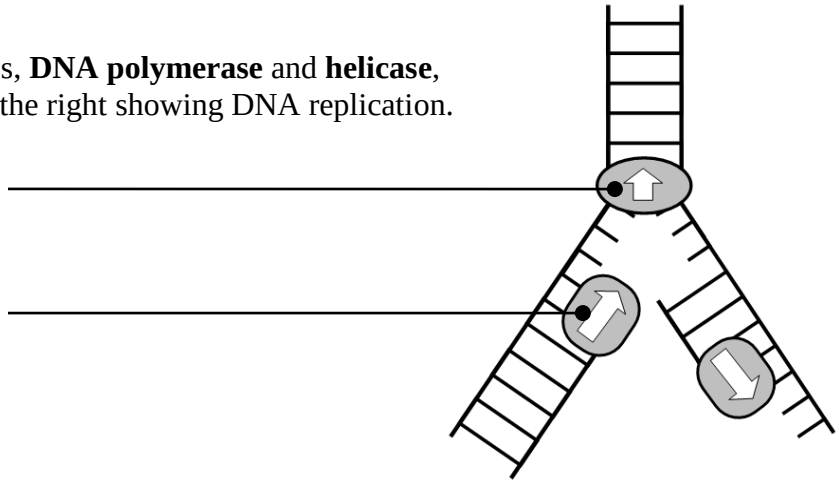
_____ Each half of the original DNA molecule acts as a template.

_____ Enzymes form new strands using complementary base pairing.

_____ An enzyme unwinds and unzips the DNA molecule.

_____ The daughter strands wind together with their parent strands forming two identical copies of DNA.

14. Label the enzymes, **DNA polymerase** and **helicase**, in the diagram to the right showing DNA replication.



15. For each of the following nitrogenous base sequences, write the **complimentary sequence** on the line provided.

a) **A – T – C – C – T – A – G – A – A – G – G – T**

_____

b) **C – G – T – T – G – C – A – G – A – A – C – T**

_____

c) **T – A – C – G – G – A – T – C – G – T – C – A**

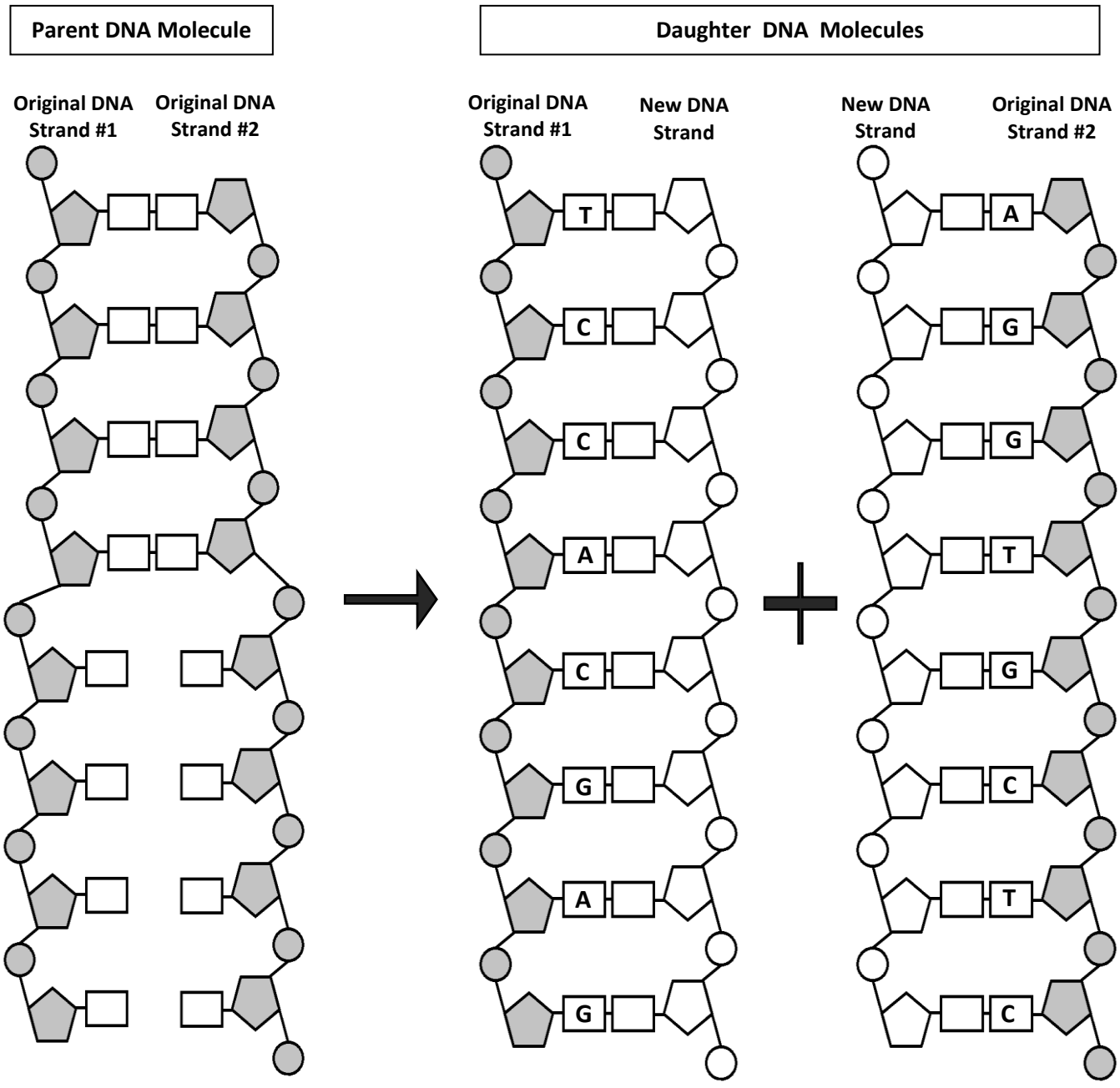
_____

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Name _____ Period _____ Date _____

DNA Structure & Replication, *Continued*

16. The diagrams below help illustrate the **semi-conservative replication of DNA**. The dark strands belong to the original DNA molecule. The white strands represent the newly formed strands. Fill in the missing complementary nitrogenous bases to complete the two daughter DNA molecules. Then, complete the parent DNA molecule to show what it looked like before replication began.



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ANSWER KEY

DNA Structure & Replication

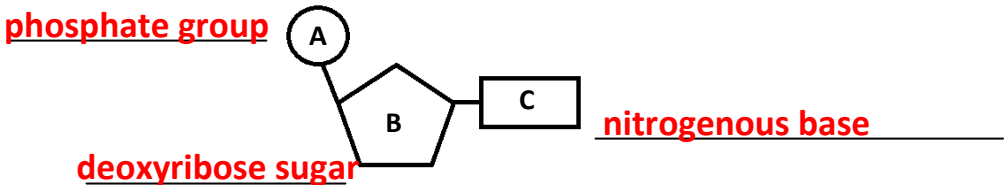
Directions: Answer each question, complete each statement, or label each diagram as indicated. You may use your textbook, notes, or other appropriate resources.

17. What is the full name of **DNA**? deoxyribonucleic acid

18. Segments of DNA which code for specific proteins are called genes.

19. DNA may coil and condense into visible structures called chromosomes.

20. DNA is a polymer of **nucleotides**. Label the parts of the nucleotide shown below.

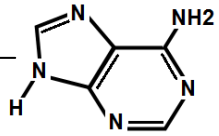


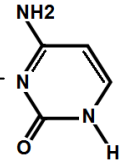
21. Four **nitrogenous bases** are found in **DNA**.

a) Which nitrogenous bases are **purines**? adenine guanine

b) Which nitrogenous bases are **pyrimidines**? cytosine thymine

22. Classify the nitrogenous bases below as either a **purine** or a **pyrimidine**.

purine 

pyrimidine 

23. Nitrogenous bases form pairs in specific combinations. Identify these pairs in the spaces below.

c) adenine pairs with thymine

d) cytosine pairs with guanine

24. What chemical **bond** holds the nitrogenous base pairs together? hydrogen bond

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ANSWER KEY

DNA Structure & Replication, *Continued*

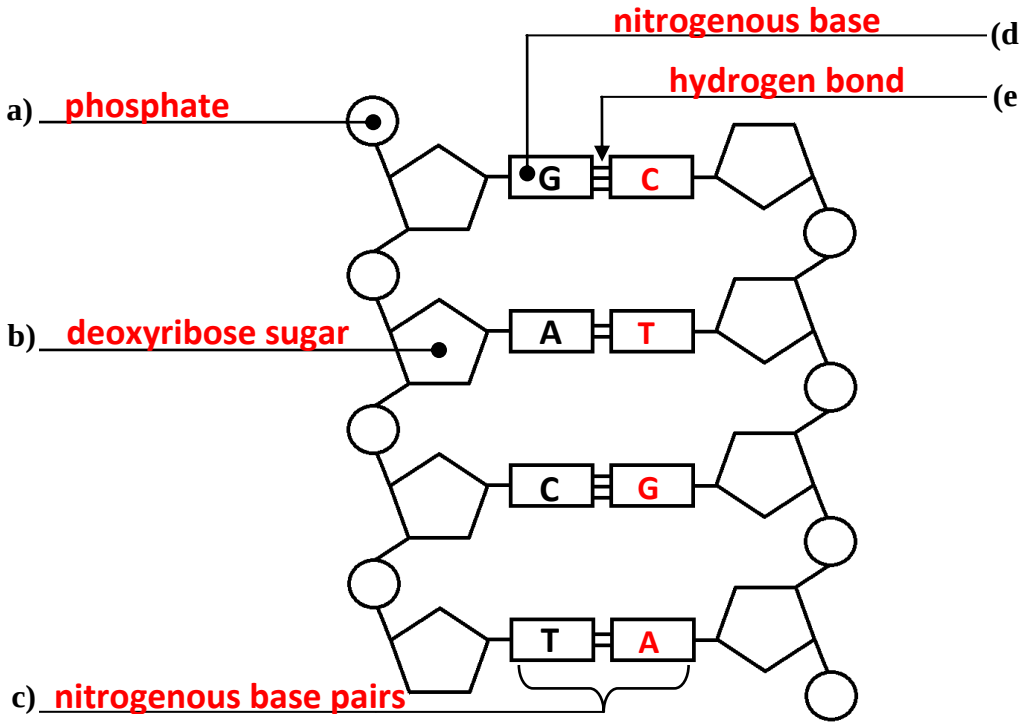
25. James Watson and Francis Crick, with the help of Rosalind Franklin, determined that DNA is in the shape of a _____double_____ _____helix_____.

26. The shape of DNA can be compared to a twisted ladder.

    a) The sides of the ladder would be the _deoxyribose sugar_ and ____phosphate____.

    b) The rungs or steps of the ladder would be the _____nitrogenous base pairs_____.

27. Label the DNA diagram shown below using the follow terms: **base pair, deoxyribose**, **hydrogen bond**, **nitrogenous base**, and **phosphate group**. Then, fill in the missing nitrogenous bases.



28. DNA undergoes **replication** before a cell divides by mitosis or meiosis.

    a) What enzyme "unzips" the DNA molecule? ____helicase____

    b) What enzyme builds the new DNA strands using the original strands as templates?

    ____DNA polymerase____

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ANSWER KEY

DNA Structure & Replication, *Continued*

29. Number the following steps of **DNA replication** in the correct order (1, 2, 3, …)

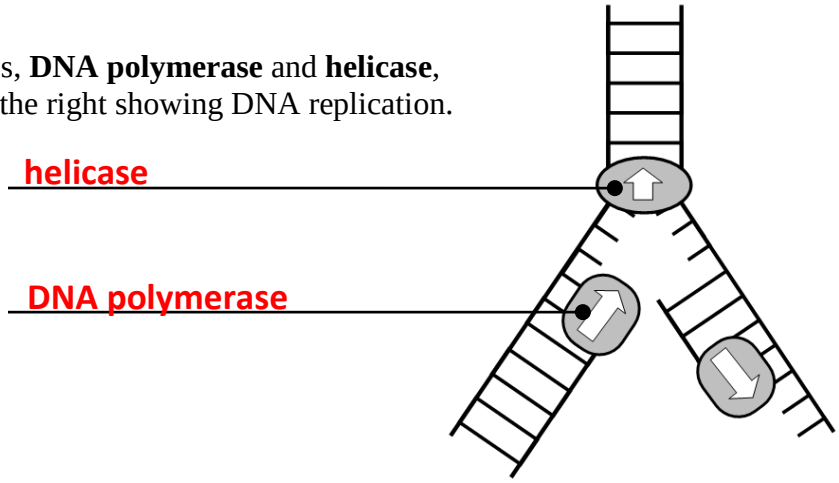
__2__ Each half of the original DNA molecule acts as a template.

__3__ Enzymes form new strands using complementary base pairing.

__1__ An enzyme unwinds and unzips the DNA molecule.

__4__ The daughter strands wind together with their parent strands forming two identical copies of DNA.

30. Label the enzymes, **DNA polymerase** and **helicase**, in the diagram to the right showing DNA replication.



31. For each of the following nitrogenous base sequences, write the **complimentary sequence** on the line provided.

a) **A – T – C – C – T – A – G – A – A – G – G – T**

T – A – G – G – A – T – C – T – T – C – C – A

b) **C – G – T – T – G – C – A – G – A – A – C – T**

G – C – A – A – C – G – T – C – T – T – G – A

c) **T – A – C – G – G – A – T – C – G – T – C – A**

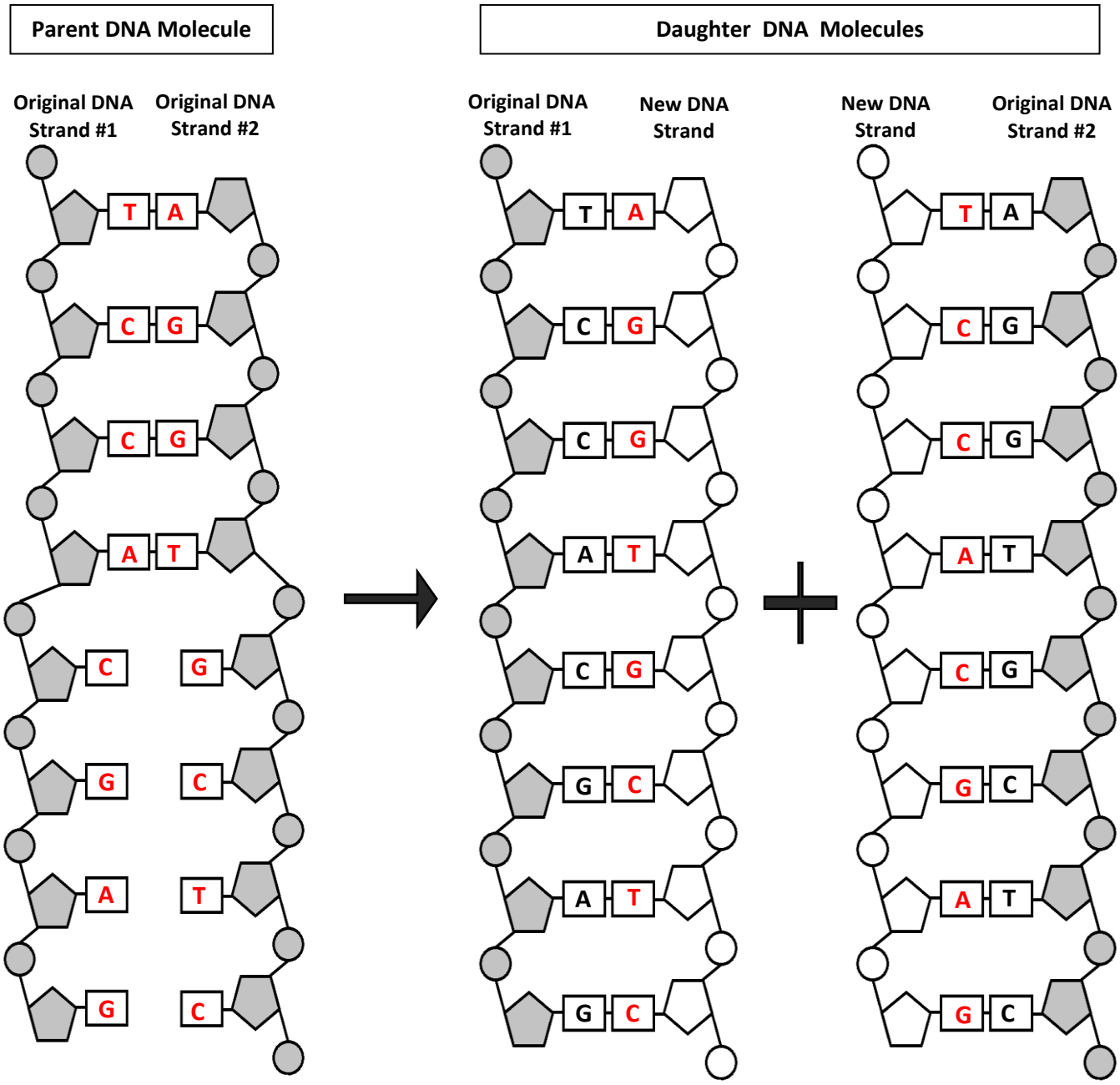
A – T – G – C – C – T – A – G – C – A – G – T

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ANSWER KEY

DNA Structure & Replication, *Continued*

32. The diagrams below help illustrate the **semi-conservative replication of DNA**. The dark strands belong to the original DNA molecule. The white strands represent the newly formed strands. Fill in the missing complementary nitrogenous bases to complete the two daughter DNA molecules. Then, complete the parent DNA molecule to show what it looked like before replication began.



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